

IT Infrastructure & Systems Portfolio

Candidate: Ramakant Joshi

Location: Copenhagen, Denmark

Education: Technical Education Copenhagen (TEC)

Contact: rajo@elev.tec.dk | 41605001 |

<https://www.linkedin.com/in/ramakant-joshi-529882127/>

Portfolio Overview

This portfolio contains documented engineering labs demonstrating my hands-on capabilities in **Systems Administration (Windows Server 2022)** environments were designed, configured, and verified by me to simulate real-world enterprise architectures.

Project 1: Enterprise Active Directory & Group Policy Deployment

Project Overview

- **Objective:** Design, deploy, and secure a centralized Windows Domain environment using VirtualBox to simulate a corporate infrastructure.
- **Core Technologies:** Windows Server 2022 Standard, Active Directory Domain Services (AD DS), DNS, DHCP, and Group Policy Management.
- **Environment Parameters:** Domain: firma.local | DC Hostname: SRV-01 | Network: 192.168.1.0/24

Implementation Details

1. **Network Baseline:** Configured Domain Controller with a static IP (192.168.1.10). Configured workstation PC1 to use the server as its preferred DNS to successfully bind the client to the firma.local tree.

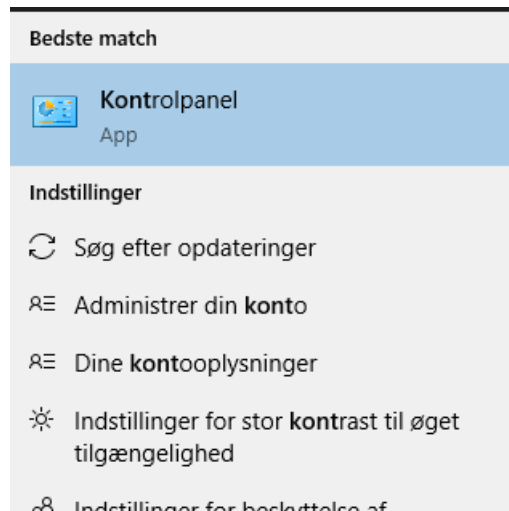
Computer name	WIN-VP422UG6P76	Last installed updates	Never
Domain	firma.local	Windows Update	Download u
		Last checked for updates	Never
Microsoft Defender Firewall	Public: On	Microsoft Defender Antivirus	Real-Time P
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC-08:00)
Ethernet	192.168.1.10, IPv6 enabled	Product ID	00454-4000
Operating system version	Microsoft Windows Server 2022 Standard Evaluation	Processors	Intel(R) Xeon
Hardware information	innotek GmbH VirtualBox	Installed memory (RAM)	5.57 GB
		Total disk space	49.39 GB

- Directory Identity Structure:** Deployed Active Directory Domain Services. Built a dedicated Organizational Unit (OU) named Elever to segregate restricted user profiles from core staff.

☐ Active Directory Certificate Services
☒ Active Directory Domain Services
☐ Active Directory Federation Services
☐ Active Directory Lightweight Directory Services
☐ Active Directory Rights Management Services
☐ Device Health Attestation
☒ DHCP Server
☒ DNS Server
☐ Fax Server
☒ File and Storage Services (1 of 12 installed)
☐ Host Guardian Service
☐ Hyper-V
☐ Network Policy and Access Services
☐ Print and Document Services
☐ Remote Access
☐ Remote Desktop Services
☐ Volume Activation Services
☐ Web Server (IIS)
☐ Windows Deployment Services
☐ Windows Server Update Services

- Security Hardening (GPO):** Created and enforced a Group Policy Object (GPO) titled Disable control panel. Tied this rule to the Elever OU to restrict access to the control panel subsystem.

Elever						
Linked Group Policy Objects		Group Policy Inheritance	Delegation			
Link Order	GPO	Enforced	Link Enabled	GPO Status	WMI Fil	
1	Disable control panel	No	Yes	Enabled	None	



Verification & Diagnostics

- **Network Connectivity:** Executed bidirectional ping sweeps between host and guest environments to verify Layer 3 connectivity.
- **Service Availability Check:** Validated that crucial authentication and name resolution ports were online using command line telnet mapping:

cmd

telnet firma.local 53

- **GPO Enforcement Proof:** Logging into the client workstation under a user account in the Elever OU and trying to open the Control Panel displays the following native security barrier:

